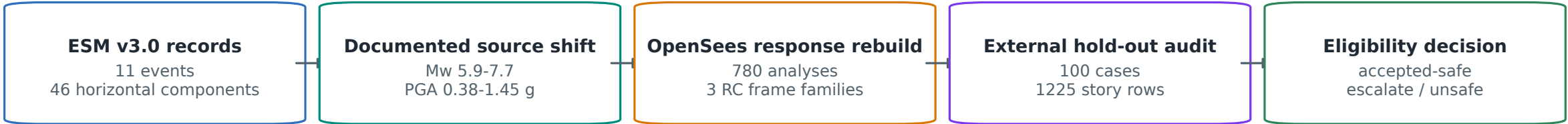


ESM external validation converts source shift into a safety eligibility audit

The evidence chain documents public strong-motion provenance, derived nonlinear response reconstruction, and the margin-controlled false-safe decision surface.

A External-source validation as a reliability evidence chain



Raw ESM accelerograms are third-party records; the manuscript reports derived response metadata, selection provenance, and reproducible analysis scripts.

B ESM source envelope

46 Components

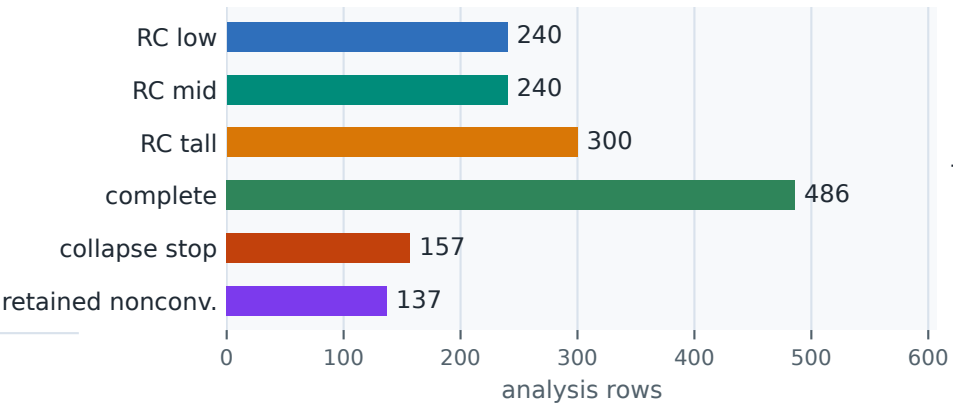
780 Response analyses

1225 Hold-out story rows

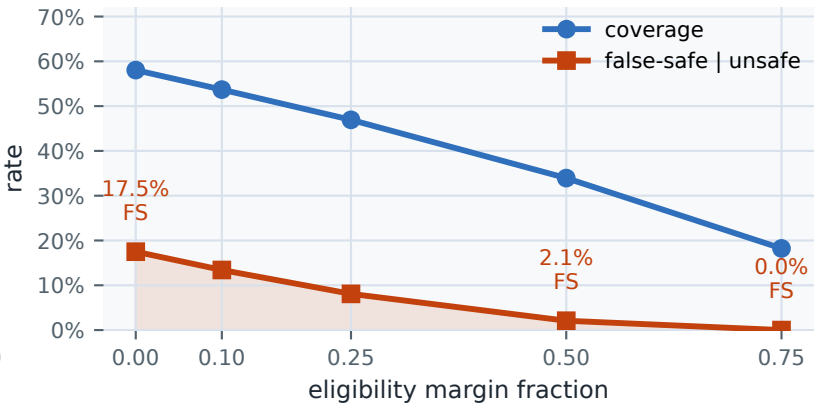
400 Unsafe rows at 2% IDR

Mw 5.9-7.7; distance 3.4-78.2 km; dt 0.005 s.

C High-fidelity response rebuild composition



D 2% IDR safety-coverage gate



E Event-tail false-safe exposure

